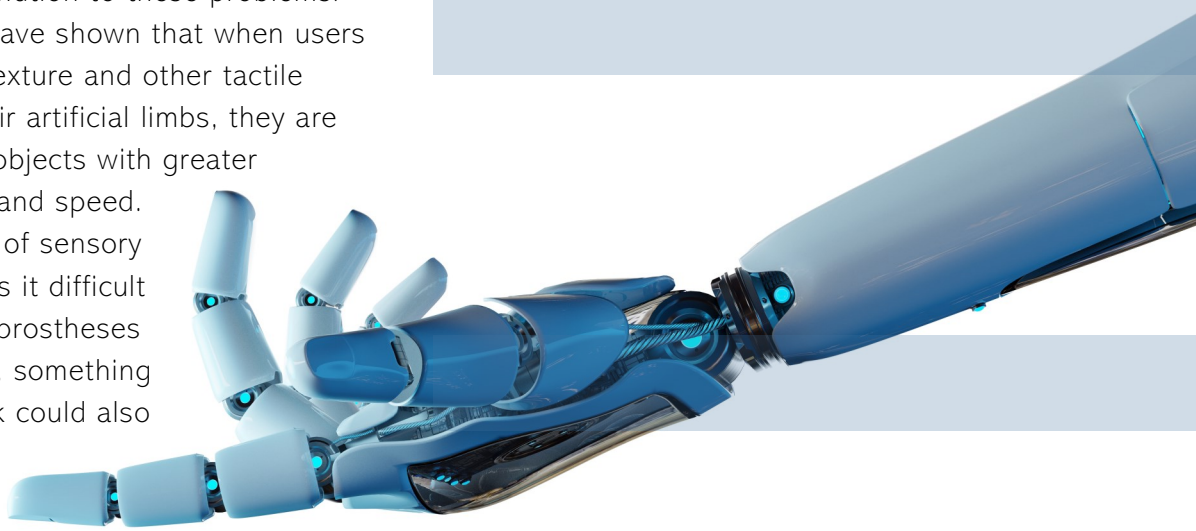
# Prosthetics and Haptic Feedback

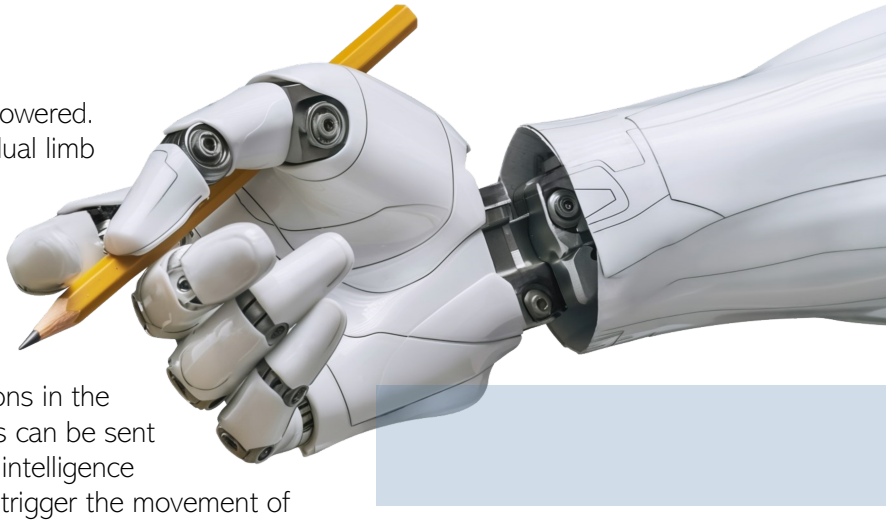


Image taken from Shutterstock.

In the past years, prostheses have transformed the lives of many individuals living with limb loss. A key factor in enabling reliable operation of these prosthetics is a closed-loop Human Machine Interface (HMI) that should involve both motor control and sensory feedback. However, most commercially produced prostheses have open-loop control, since although they can largely mimic the mechanical functions of human limbs, they lack tactile feedback, which is a major roadblock in preventing users from efficiently handling their prosthetics. [1] Many have reported that using them to perform daily tasks can be incredibly mentally taxing, since they must rely heavily on visual feedback to constantly monitor their prosthetic limb. This has led some to abandon their device altogether. Adding haptic feedback to prostheses and adopting a fully closed-loop HMI could be the solution to these problems. Many new studies have shown that when users can feel pressure, texture and other tactile sensations from their artificial limbs, they are able to manipulate objects with greater precision, dexterity and speed. Furthermore, a lack of sensory feedback also makes it difficult for users to accept prostheses as their 'own limbs', something that haptic feedback could also help change.

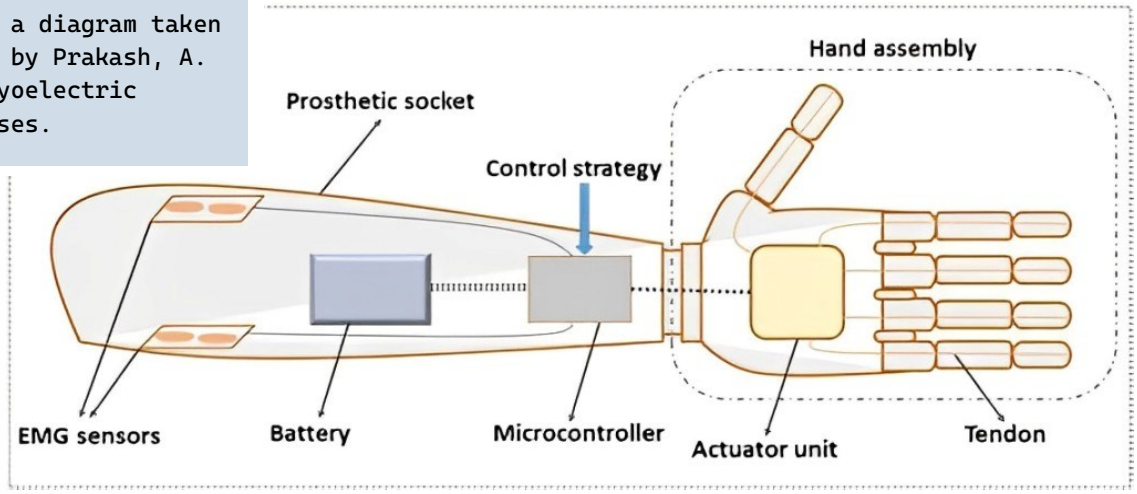
**"Our researchers are addressing the loss of tactile sensations, which is currently a major roadblock in preventing upper-limb-absent people from multitasking or using the full dexterity of their prosthetic hands" – Florida Atlantic's Stella Batalama in an interview with *Science Daily***





There are three main ways that prostheses are powered. Body powered prosthetics involve using the residual limb to directly move the artificial limb using a cable. Motor powered prosthetics have buttons and switches which control the artificial limb. And, finally, myoelectric powered prosthetics contain electromyogram (EMG) sensors in the socket of the artificial limb which use electrodes to detect electrical signals from muscle contractions in the residual limb. Powered by a battery, these signals can be sent to a microcontroller, which can then use artificial intelligence algorithms to identify patterns in the signals and trigger the movement of motors in the prosthetic, hence allowing it to perform different mechanical functions. This method is usually preferred over the other two since it often looks more natural, is more comfortable and offers finer and more intuitive control of the prosthetic.

**On the right is a diagram taken from an article by Prakash, A. of a standard myoelectric powered prostheses.**

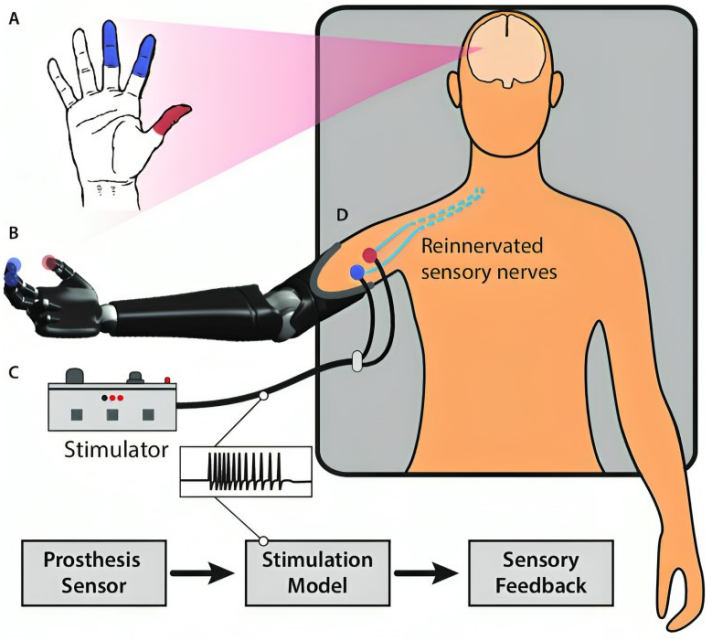




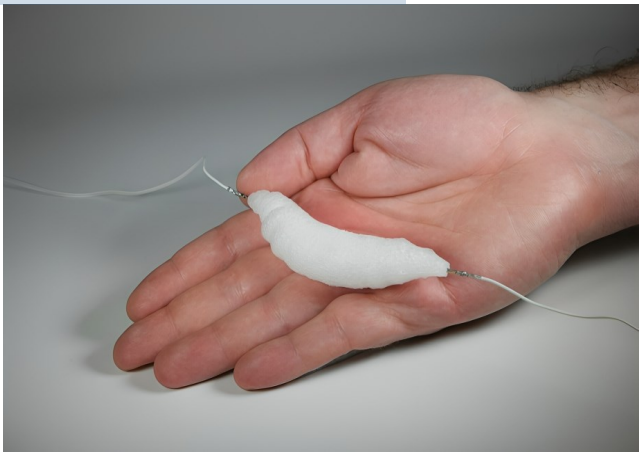

The first step in introducing haptic feedback to prostheses is to find where on the body this feedback can be administered. Scientists have found that amputees and people born without a limb may often continue to feel sensations in their missing limb. This is because the nerves that originally sent signals from the missing limb to the brain are still present in the residual limb and stimulating them can trick the person into thinking that the sensations are coming from the missing limb. This is called having a phantom limb. Hence, mapping these nerves to locations on the residual limb could reveal where haptic feedback should be applied.

Haptic feedback can be administered in several different ways. Some of these involve invasive approaches that involve directly interfacing with nerves. Although somewhat successful, it can be quite dangerous. Non-invasive approaches are more common, and perhaps the most popular of these is vibrotactile by using vibrotactile actuators. This method has been achieved in, for example, myoelectric-controlled hands. It involves incorporating different sensors that are attached to the hand. They can sense different forces, directions, textures and so on of objects that the prosthetic interacts with. The information detected by these sensors can change the stimulation parameters, such as stimulation position, frequency, amplitude and duration, of actuators attached to the user's residual limb that will then vibrate in proportion to the haptic feedback. This allows users to feel different tactile sensations such as pressure and an overall sense of touch.

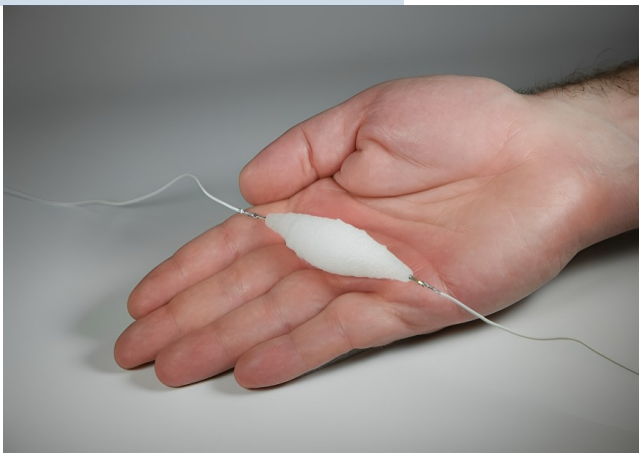
On the right is a diagram taken from an article written by Luke E. Osborn of a standard haptic feedback system for prostheses. It includes sensors on the prosthetic, a simulation model that interprets information from the sensors and connections to sensory nerves that enable sensory feedback.




## Before activation

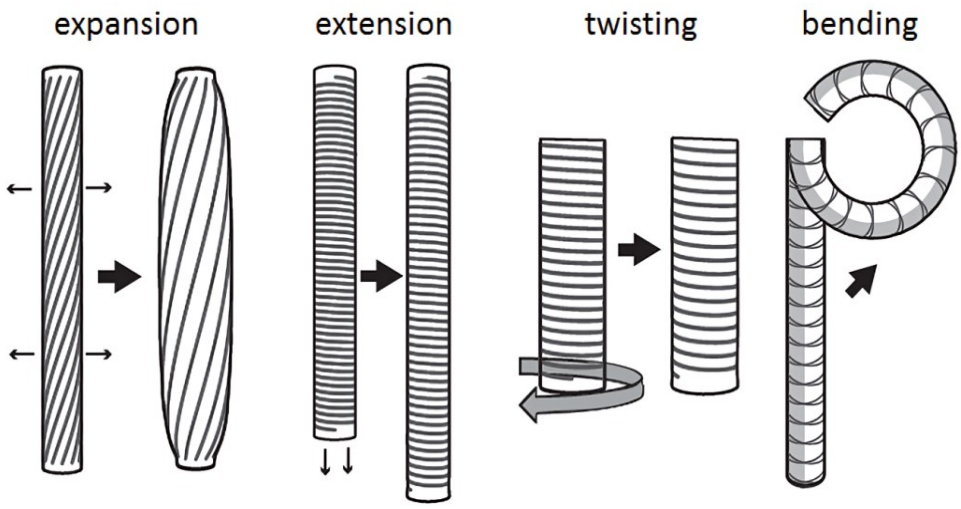


## After activation



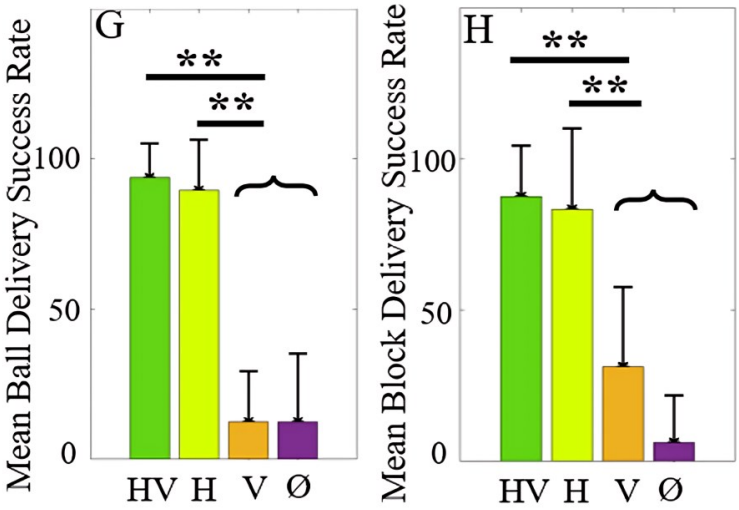
More recently, soft robotic actuators have also been used instead of vibrotactile actuators to administer haptic feedback. Similar sensors are used, and similar parameters are measured in both methods, but soft robotic actuators are made of flexible elastomers that imitate biological tissues, making them more comfortable and realistic. They usually involve some sort of chamber into which air can be pumped or released, causing it to inflate or deflate. This allows them to more versatile than vibrotactile actuators since not only can they imitate vibrations by inflating and deflating rapidly, but they could also change shape by inflating and deflating in certain places. This might allow for a wider range of more precise tactile sensations.

The diagrams on the left are taken from an article written by Miriyev, A. and show a soft robotic actuator that curves and elongates after being activated. An actuator like this could, for example, provide sensory feedback in the form of tension.




The diagrams on the left are taken from an article written on the Soft Robotics Toolkit website and demonstrate a range of different operations that a soft robotic actuator could perform. These could all be used to provide different forms of sensory feedback.

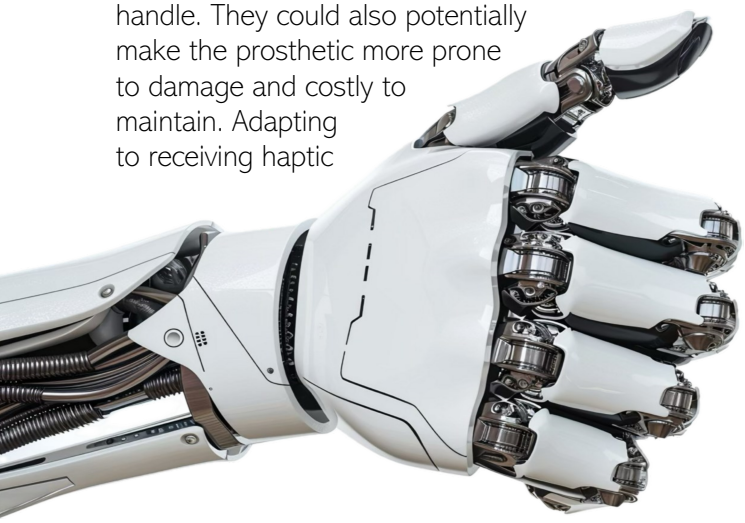
Multiple studies have shown that individuals with absent limbs find it easier to carry out certain tasks with haptic feedback compared to without. For example, one study published in 2022 in *Nature Scientific Reports* investigated the benefits of haptic feedback for upper limb prostheses and involved the use of soft robotic actuators worn in an armband wrapped around the upper arm to help grasp and delivery two balls simultaneously. When haptic feedback was involved the grasping success rate was 90.6% ± 14.0% and the delivery success rate was 86.3% ± 21.7% but these values plummeted to 21.9% ± 21.6%, and 9.4% ± 19.0% respectively when haptic feedback was disabled. Many other studies show similar results, such another 2022 study published in the journal *Medecine*. It described how a vibrotactile feedback system improved balance, gait and walking speed for a below-knee amputee. All of these cases are evidence for the potential haptic feedback has for significantly improving the quality of life of those living with missing limbs. On the other hand, one obstacle that this new technology might face is that phantom limb sensations are not constant and usually only last for a period of time. Some may not even experience it at all. Finding where to administer haptic feedback could therefore be a problem when the sensation is not present.




**Graphs from a study written by Abd M.A. that shows the benefit of haptic feedback over visual feedback alone. H means haptic feedback only, V means visual feedback only, HV means both and Ø means neither.**



In my opinion, whilst haptic feedback in prosthetics offer significant benefits, it could also potentially lead to certain challenges. First of all, it involves incorporating more complex technology such as vibrotactile actuators, soft robotic actuators and sensors into the prosthetic that would increase its cost. This could put those with a lower budget at a disadvantage. These components would also add extra weight to the artificial limb, which could be tiring for the user to handle. They could also potentially make the prosthetic more prone to damage and costly to maintain. Adapting to receiving haptic feedback could also be difficult and time-consuming. Additionally, actuators may cause the user discomfort, such as irritation to the skin.



However, I believe that the advantages haptic feedback has to offer greatly outweigh the potential disadvantages. Although the extra components might make the prosthetic more expensive, as it becomes more widely manufactured, its cost will surely decrease. Making these components more durable and less heavy could also be goals for the future and as the field of technology rapidly advancements, solutions will arise. As for possible discomfort, feedback from the users could be used to better the prosthetics and continuously alter them until they are comfortable to wear.

As discussed previously, living with an absent limb can be incredibly difficult. It creates barriers to occupational and social participation as well as limiting the freedom and capability to do daily activities. Haptic feedback has proved to be essential in giving these individuals a much better quality of life.

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